

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8493**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2021600049
Course Section:	1				

L_29_506_986_L_1505_38331



Name of the candidate. Mayank Ramji Ravariya

Candidate's UID number:

2	0	2	2	6	0	0	0	4	9
---	---	---	---	---	---	---	---	---	---

Examination class room number:

4	0	1
---	---	---

Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTech MGT
(For example FY MCA/FY BTECH)

SEMESTER : X

PROGRAM: CSE-AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8-10-25 Wednesday

COURSE NAME: Deep Learning

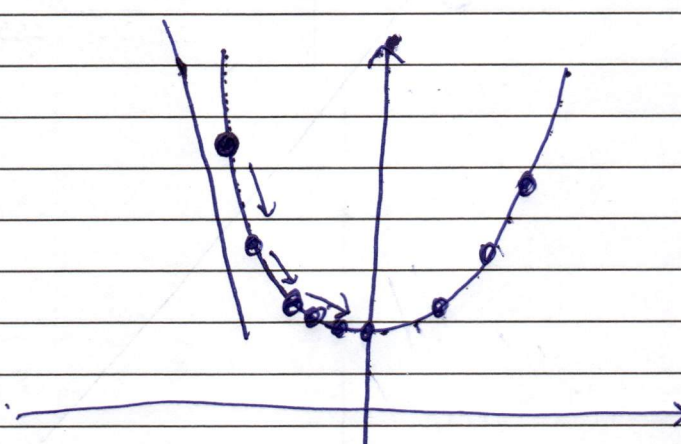
(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



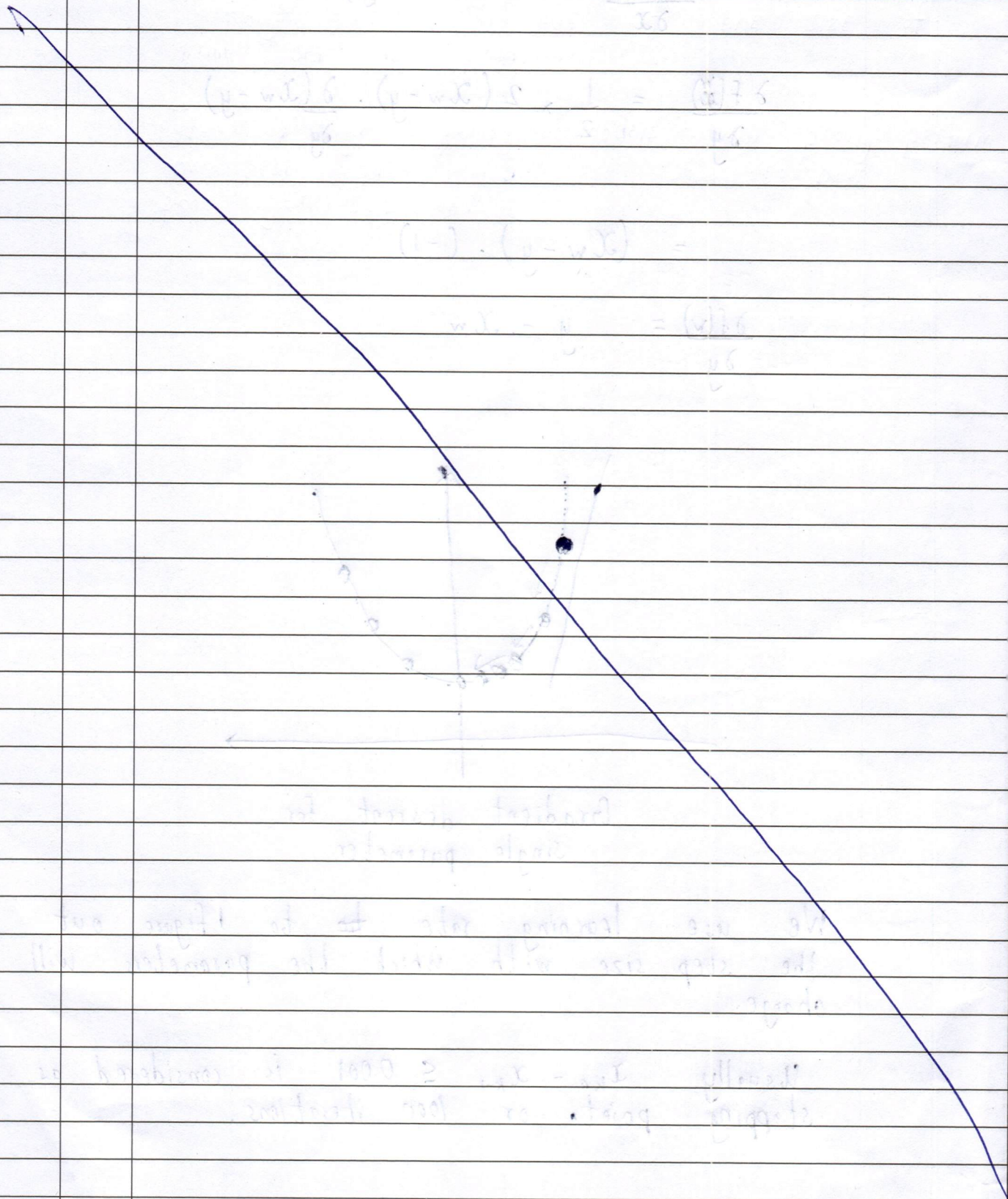
Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q.1a	
	→	$\frac{\partial f(w)}{\partial x} = \frac{1}{2} \times 2(xw - y) \cdot \frac{\partial (xw - y)}{\partial x}$ $= (xw - y) \cdot w$ $\therefore \frac{\partial f(w)}{\partial x} = w \cdot (xw - y)$ $\frac{\partial f(w)}{\partial y} = \frac{1}{2} \times 2(xw - y) \cdot \frac{\partial (xw - y)}{\partial y}$ $= (xw - y) \cdot (-1)$ $\therefore \frac{\partial f(w)}{\partial y} = y - xw$
		
		Gradient descent for single parameter
		We use learning rate to to figure out the step size with which the parameters will change.
		Usually $x_k - x_{k-1} \leq 0.001$ is considered as stopping point or 1000 iterations.



Space
For
Marks

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)



Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

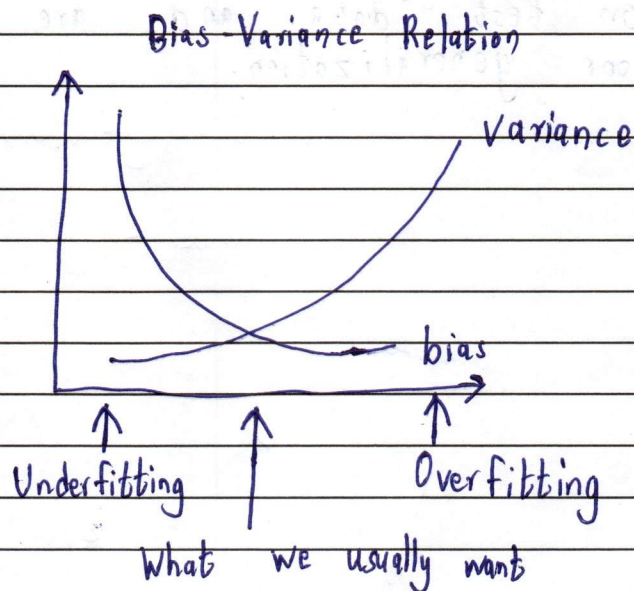
Q.1b

→

A model is said to have high bias when it is too simple and doesn't correctly predict the classes.

~~A~~ Variance is the measure of deviation of model output with ~~sa~~ small changes to input.

~~A model is said to have high variance~~
Overfitting occurs when the model has high variance.



$$\text{Mean Square error (MSE)} = \frac{1}{n} \sum (y - \hat{y})^2$$

Model capacity is how well can the model fit a variety of functions.

Simple models are Linear regression are said to have low capacity since it just fits linear separable data.

Models like CNN, Transformer, Deep Feedforward Networks are said to have high capacity as these can fit and be used for a variety of usecases.

Space
For
MarksQuestion
No.

START WRITING HERE

(Begin answer for each question on a new page)

Generalization ~~is the ab~~ means that the model learns ~~to~~ from training data and has good performance on test data.

High Capacity models are prone to overfitting as they have a large number of parameters which can easily ~~remember~~ remember instead of learning the train set if the size of train set is small.

Thus High capacity models will poorly perform on test data and are said to have poor generalization.

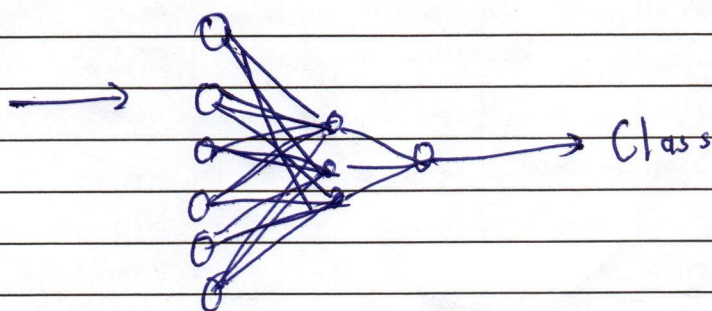
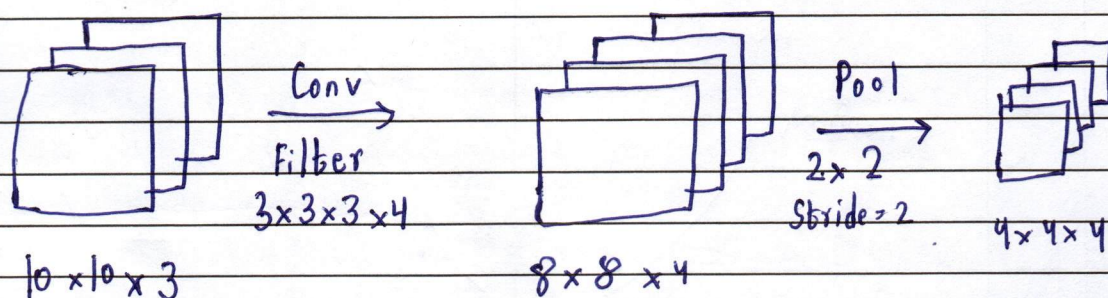
Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Q.2a

→ Formula for Output Image size

$$= \left\lfloor \frac{N - F + 2P}{S} + 1 \right\rfloor$$

where N = Input image dimension F = Filter size P = Padding S = Stride

CNNs help us to perform deep learning on images where ~~sp~~ spacial context matters and can be taken advantage of.

Imagine if we had 1000×1000 pixel image, input layer would alone be 1000000 neurons.

This would consume too much memory.

By using CNNs we can convert it into 32×32 pixels and then pass it to Fully connected layers.



START WRITING HERE

(Begin answer for each question on a new page)

Space
For
MarksQuestion
No.

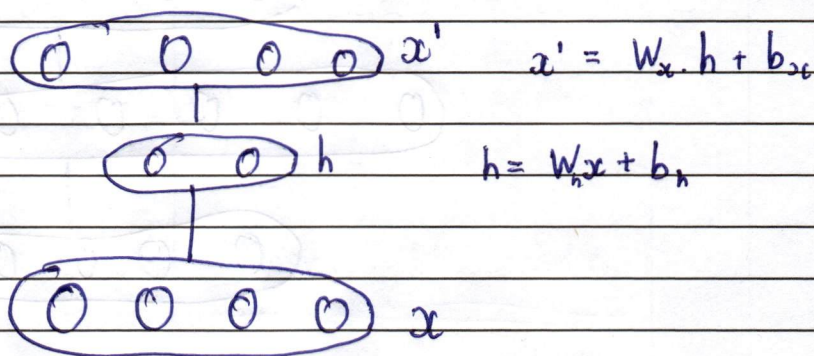
We can train CNN on similar category of images to learn feature maps and then perform transfer learning to quickly apply it in another use case.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Q.2b

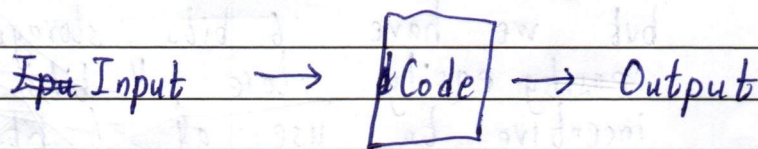
→



Linear Undercomplete Autoencoder

Undercomplete autoencoders are used to compress data and then ~~res~~ ~~res~~ reconstruct it.

We use non-linear activation functions to ensure that ~~the~~ the latent space or the "code" layer learns features from input x .



$$\dim(h) < \dim(x)$$

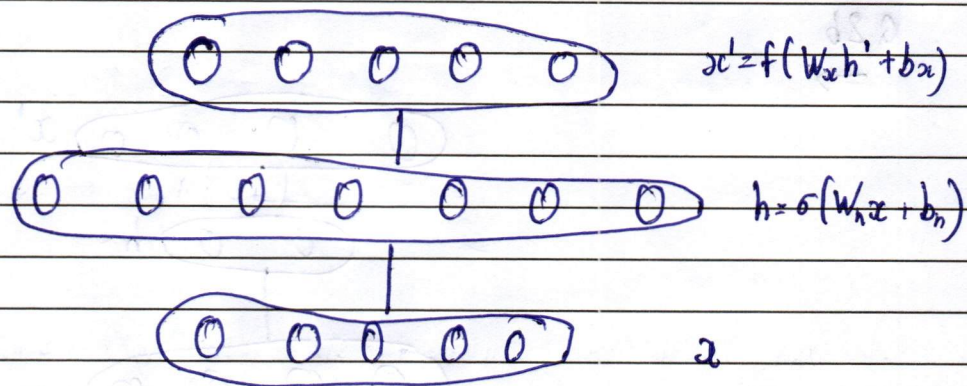
~~Autoen~~ Under complete autoencoder in essence, reduces the feature dimensions to only keep important features (Dimensionality Reduction)

Without non-linear activation functions, the model will perform equivalent to PCA.

Space
For
MarksQuestion
No.

START WRITING HERE

(Begin answer for each question on a new page)



$$\dim(h) \geq \dim(x)$$

Overcomplete Autoencoder

When the dimensions of hidden layer or "code" layer are greater than x , the model can store the x instead of reconstructing itself. This and then output the same x . This is known as identity function.

Its like we want to store 4 bits but we have 6 bits storage. Model can easily store 4 bits & not having any incentive to use all 6 bits.

Thus we add additional constraints by regularization to prevent model from learning the identity function.

We can also physically add constraints by ~~by turning~~ like in case of Sparse encoders.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

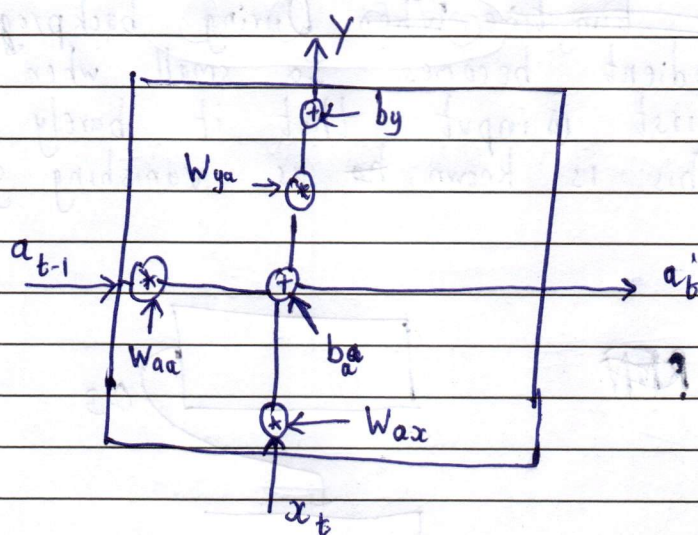
Q3a.

→

RNN.

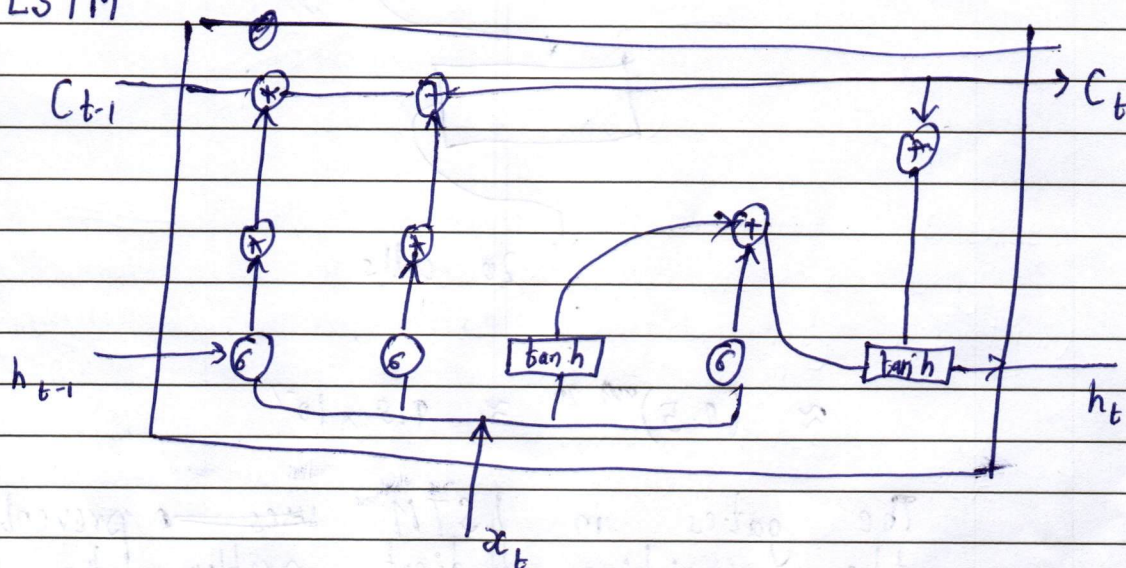
$$a_t = f(W_{ax} \cdot x_t + W_{aa} \cdot a_{t-1} + b_a)$$

$$Y = \text{softmax}(W_{ya} \cdot a_t + b_y)$$



RNN architecture

LSTM



$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

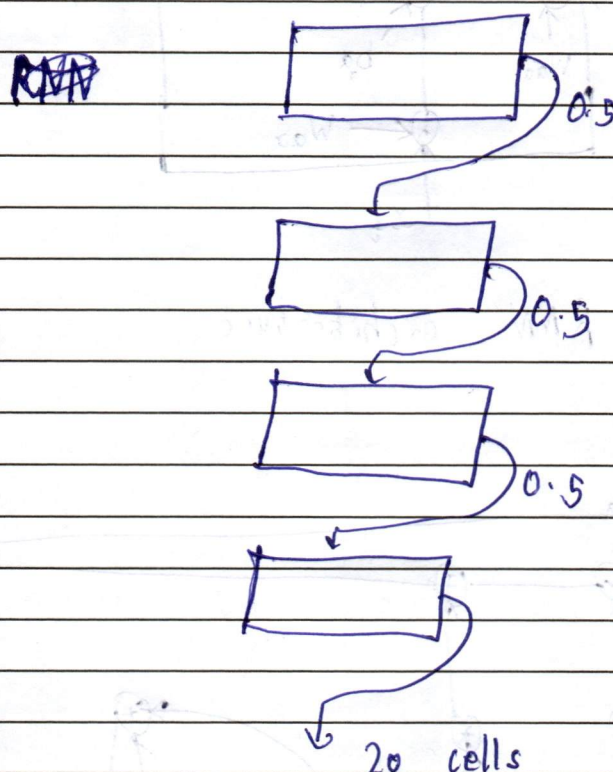
Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

$$C_t = i_t * \tilde{C}_t + f_t * C_{t-1}$$

$$h_t = o_t * \tanh(C_t)$$

When the weights are small, i.e. less than 1.
~~Over time~~ During backpropagation the
gradient becomes so small when we reach the
first input that it barely updates.
This is known as vanishing gradient problem.



$$\approx (0.5)^{20} \approx 9.5 \times 10^{-7}$$

The gates in LSTM ~~uses~~ prevents the vanishing gradient problem by only allowing percentage of input to update output.

If we use ReLU activation function, the derivative is 1 and vanishing / exploding gradient problem does not occur.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Q2b.

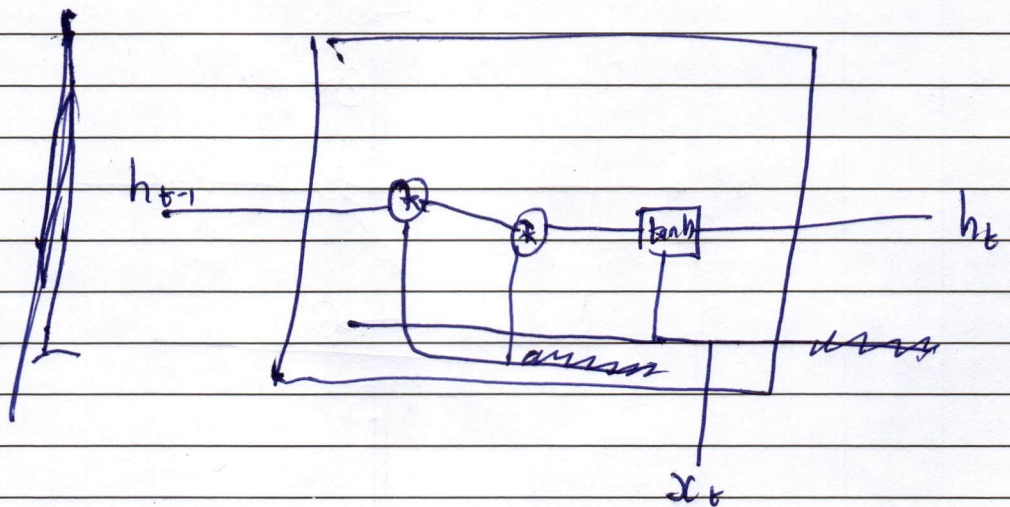
$$\rightarrow u_t = \sigma(W_u[h_{t-1}, x_t] + b_u)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r)$$

$$\tilde{C}_t = \tanh(W_c[r_t * h_{t-1}, x_t] + b_c)$$

$$h_t = u_t * \tilde{C}_t + (1 - u_t) h_{t-1}$$

GRU Equations & Unit



GRU is a streamlined version of LSTM and gives near about good results as LSTM.

It is useful to be used in resource-intensive environment such as hardware applications as its performance is significantly better than RNN and doesn't consume ~~much~~ as much memory as LSTM.

GRU has only 2 gates, LSTM has 3.

Less no. of parameters also mean less chances of overfitting which means GRUs can have better generalization.

[illegible]

